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# Microwave assisted organic synthesis- A boon for green chemistry & environment

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**Abstract:** Microwave-assisted organic synthesis is considered to be a promising green chemical approach because it reduces reaction time from days or hours to minutes or even seconds, and has many other advantages. Green chemistry reduces pollution at its source by minimizing or eliminating the hazards of chemical feedstocks, reagents, solvents, and products. If a technology reduces or eliminates the hazardous chemicals used to clean up environmental contaminants, this technology would qualify as a green chemistry technology.Science is making chemical processes more sustainable by reducing the amount of compounds used and reducing and/or eliminating the amount of toxic substances we produce. Researchers have made important progress in Green Chemistry in recent years.Green chemistry is the design of chemical products and processes that reduce or eliminate the generation of hazardous substances.In modern chemistry, principles are the constituents of a substance, specifically those that produce a certain quality or effect in the substance, such as a bitter principle, which is any one of the numerous compounds having a bitter taste. The idea of chemical principles developed out of the classical elements.

**Key words: Stoichiometric reagent,derivatization,biological pathways.**

## INTRODUCTION

### Green chemistry

Green chemistry, also called sustainable chemistry, is an area of chemistry and chemical engineering focused on the designing of products and processes that minimize the use and generation of hazardous substances.[1] Whereas environmental chemistry focuses on the effects of polluting chemicals on nature, green chemistry focuses on technological approaches to preventing pollution and reducing consumption of nonrenewable resources.[2,3,4,5,6,7]

Green chemistry emerged from a variety of existing ideas and research efforts (such as atom economy and catalysis) in the period leading up to the 1990s, in the context of increasing attention to problems of chemical pollution and resource depletion. The development of green chemistry in Europe and the United States was linked to a shift in environmental problem-solving strategies: a movement from command and control regulation and mandated reduction of industrial emissions at the "end of the pipe," toward the active prevention of pollution through the innovative design of production technologies themselves.

**The twelve principles of green chemistry are:**

It is better to prevent waste than to treat or clean up waste after it is formed.

Synthetic methods should be designed to maximize the incorporation of all materials used in the process into the final product.

Wherever practicable, synthetic methodologies should be designed to use and generate substances that

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possess little or no toxicity to human health and the environment.

Chemical products should be designed to preserve efficacy of function while reducing toxicity.

The use of auxiliary substances (e.g. solvents, separation agents, etc.) should be made unnecessary wherever possible and innocuous when used.

Energy requirements should be recognized for their environmental and economic impacts and should be minimized. Synthetic methods should be conducted at ambient temperature and pressure.

A raw material or feedstock should be renewable rather than depleting wherever technically and economically practicable.

Reduce derivatives – Unnecessary derivatization (blocking group, protection/deprotection, temporary modification) should be avoided whenever possible.

Catalytic reagents (as selective as possible) are superior to stoichiometric reagents.

Chemical products should be designed so that at the end of their function they do not persist in the environment and break down into innocuous degradation products.

Analytical methodologies need to be further developed to allow for real-time, in-process monitoring and control prior to the formation of hazardous substances.

Substances and the form of a substance used in a chemical process should be chosen to minimize potential for chemical accidents, including releases, explosions, and fires.

**Benefits of Green Chemistry Environment:**

Many chemicals end up in the environment by intentional release during use (e.g. pesticides), by unintended releases (including emissions during manufacturing), or by disposal. Green chemicals either degrade to innocuous products or are recovered for further use.

Plants and animals suffer less harm from toxic chemicals in the environment.

Lower potential for global warming, ozone depletion, and smog formation.

Less chemical disruption of ecosystems.

Less use of landfills, especially hazardous waste landfills.

**Human health:**

Cleaner air: Less release of hazardous chemicals to air leading to less damage to lungs.

Cleaner water: less release of hazardous chemical wastes to water leading to cleaner drinking and recreational water.

Increased safety for workers in the chemical industry; less use of toxic materials; less personal protective equipment required; less potential for accidents (e.g., fires or explosions)

Safer consumer products of all types: new, safer products will become available for purchase; some products (e.g., drugs) will be made with less waste; some products (i.e., pesticides, cleaning products) will be replacements for less safe products

Safer food: elimination of persistent toxic chemicals that can enter the food chain; safer pesticides that are toxic only to specific pests and degrade rapidly after use.

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